

Summary

AI Engineer with an M.S. in Electrical Engineering, focused on backend services, real-time control platforms, AI data workflows, and system integration. Experienced with FastAPI, WebSocket, React / TypeScript, Next.js, Docker, Linux, C# / .NET, MS SQL, and computer-vision / AI workflows including PatchCore; delivered systems connecting robot, vision, data, and policy services.

Key Results

3-4x imitation-learning / GROOT-VLA training-data throughput	70% → 90% cable-insertion success in 10 trials
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Experience

AMTRAN TECHNOLOGY CO., LTD. | AI Robotics Engineer May 2025 - Present | New Taipei, Taiwan

- Owned software delivery from requirements and system architecture through Docker deployment, testing, on-site integration, and maintenance for real-time AI and robotics systems.
- Built a Next.js / React Flow / FastAPI / WebSocket workflow and data-recording platform that automated task execution and captured actions, images, robot state/actions, and timestamps, reaching 3-4x manual throughput and reducing manual staffing needs by two operators.
- Developed a React / TypeScript / FastAPI / ROS 2 web control and monitoring platform with real-time feedback, video, system status, control commands, and Docker / Nginx deployment.
- Developed a PatchCore anomaly-detection API server and PyQt interface with ROI extraction, resize/crop/normalize/clip preprocessing, template replacement, and anomaly-score evaluation.

ISCOM | Software Engineer Intern Feb 2022 - Jun 2022 | Kaohsiung, Taiwan

- Participated in backend API development, frontend integration, and web feature implementation using C#, ASP.NET MVC, JavaScript, jQuery, MS SQL, and Git.

Skills

Backend API & Services	FastAPI, Flask, ASP.NET MVC, WebSocket, REST APIs, MS SQL, PostgreSQL / SQLite
Frontend & Workflow Platforms	React, TypeScript, Next.js, React Flow, JavaScript, jQuery
AI / ML & Computer Vision	PatchCore anomaly detection, PyQt, PyTorch, OpenCV, YOLO OBB, AprilTag, Isaac GROOT
Deployment & Systems	Docker, Nginx, Linux, Git, Python, C++, C# / .NET, API and data-pipeline integration

Education & Honors

- M.S. in Electrical Engineering, National Taiwan Normal University | Sep 2022 - Jul 2024
- B.S. in Computer and Communication, National Pingtung University | Sep 2018 - Jun 2022
- Honors: RHCSA; FIRA Roboworld Cup First Place; National Technical Creativity Competition Third Place